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Photochemical hydrocarbon fuel regeneration: Hydrogen-rich fuel from CO_2

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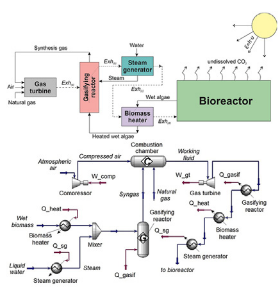
HIGHLIGHTS

- Photochemical hydrocarbon fuel regeneration concept based on CO_2 capture.
- 1 kg of captured CO_2 gives 1.88 kg of algae biomass (*Chlorella vulgaris*).
- 10% of captured CO_2 from flue gas after methane combustion gives up to 30% of fuel regeneration.
- If 36% of CO_2 is captured from flue gases, the degree of fuel regeneration is about 1.0
- In a suggested concept, the heat recuperation degree is 90% at exhaust gas temperature of 600 °C.

GRAPHICAL ABSTRACT



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ABSTRACT

This paper presents the concept of photochemical hydrocarbon fuel regeneration by CO_2 transformation to hydrogen-rich fuel. The first stage of this system is CO_2 capture and algae (*Chlorella vulgaris*) growing. The second stage is the gasification of algae biomass to produce hydrogen-rich gas and its combustion. To compile the heat and mass balance, the thermodynamic analysis was performed under various operating parameters: temperature 400–800 °C, pressure 1–10 bar 1 kg of biomass was gasified with 1.2 kg of water. The heat of combustion of hydrogen-rich gas after gasification is up to 43% higher than the heat of combustion of initial biomass. The fuel regeneration degree is up to 0.9 when 30% of CO_2 is captured by water and proceded by algae. Moreover, the analyzed photochemical fuel regeneration system allows heat recuperation. The heat regeneration degree is calculated and the maximum value is about 0.9 is reached at 600 °C.

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Introduction

Sustainable development of the world economy is inextricably linked with energy. At the same time, in the last century, fossil fuels have played a leading role in the energy sector. In the 21st century, despite advances in alternative and renewable energy sources, the share of fossil fuels in the global energy balance (according to World Energy Balance Highlights 2020 Edition presented by the International Energy Agency (IEA)) is about 64% [1].

The use of hydrocarbon fuels leads to significant emissions of carbon dioxide into the atmosphere. Most experts reasonably suppose that an increase in ambient temperature is associated with an increase in CO_2 emissions by the world industry. To minimize CO_2 emissions, different countries adopt roadmaps for decarbonizing their energy and industry. The European Union plans to transfer energy with zero CO_2 emissions by 2050 [2]. China plans to abandon coal-fired power plants, which are one of the main sources of CO_2 emission [3]. Japan and the United States also presented their programs for decarbonization in the next dozen years [4].

The development of carbon-free or low-carbon energy goes hand in hand with the development of hydrogen energy, where hydrogen or hydrogen-enriched gas of various compositions are used as fuel. Despite the advances in alternative fuels, the recent sharp rise in hydrocarbon prices shows that global demand for hydrocarbons will be stable for many years. That is why the issues of reducing CO_2 emissions at existing fuel-consuming plants are of great importance.

Scientists and engineers around the world are proposing various ways to reduce CO_2 emissions. Such methods can be divided into two groups: post-combustion and pre-combustion carbon capture. The post-combustion CO_2 capture systems include: amine-based [5,6], carbonate-based [7,8], aqueous ammonia [9,10], membranes [11,12], biology-based [13,14]. The pre-combustion systems are including carbon extraction from hydrocarbon fuel via various chemical process [13]: methane cracking, coal gasification, etc.

The biology-based post-combustion CO_2 capture systems can be highlighted because these are one of the most environmentally friendly processes. In these systems, CO_2 is processed through the process of photosynthesis. Photosynthesis is the main biological process of converting CO_2 into oxygen and complex carbon-containing substances. The conversion of CO_2 through photosynthesis can be divided into two types: the first is a direct conversion using terrestrial plants (trees, grass, etc.). The second is the dissolution of CO_2 in water and the processing of the dissolved CO_2 by algae.

The biology-based CO_2 capture systems based on the photosynthesis process can be called the photochemical CO_2 capture systems or photochemical method. The photochemical methods of CO_2 capturing have a number of undeniable advantages: relatively low cost, scalability, and environmentally friendly. Algae, in contrast to terrestrial plants, have a number of advantages for the photochemical systems, the main one of which is a high growth rate [15,16]. Bhola V. et al. [15] showed that microalgae can capture carbon dioxide up to 50 times higher than terrestrial plants. The publications on CO_2 capture by various algae are diverse.

Bhola V. et al. [15] experimentally studied the photochemical CO_2 capture systems based on algae such as Chlorella Vulgaris and Nannochloropsis gaditana. The authors found that algae of the types Chlorella, Scenedesmus, Spirulina, Nannochloropsis, and Chlorococcum are characterized not only by large and active growth but also by high tolerance to environmental conditions and CO_2 concentrations. The authors reported that 1 kg of dry biomass accounts for 1.88 kg of carbon dioxide absorbed.

Alami et al. presented the results of the investigation of CO_2 capture from flue gas by algae [17]. The authors showed that flue gases after fuel-consuming equipment are perfect as a source of CO_2 for microalgae. Under the right cultivation conditions, the degree of carbon dioxide capture by algae can reach 99%, and the slow supply of flue gases causes an increase in the growth rate of algae.

Depra et al. investigated a CO_2 capture system based on photobioreactors and determined the role of the carbon dioxide loads in the carbon footprint [18]. The authors experimentally determined the efficiency of CO_2 capture by algae *Scenedesmus Obliquus* in an artificial photobioreactor. They reported that the saturation of CO_2 in water of 384.9 kg/m^3/day ensures the optimal CO_2 capture and crop survival. For these conditions the maximum biomass output was 0.36 kg/m^3/day, the CO_2 conversion rate was 0.44 and the oxygen release rate was 0.33. Under such conditions, the maximum efficiency of CO_2 capture reached 30.76%.

The composition of algae after photosynthesis includes carbohydrates, proteins, fats, and other organic substances. At the same time, despite the difference in the composition of various algae, they mainly consist of carbon, oxygen, hydrogen, nitrogen. In other words, algae are typical biomass. Moreover, the gasification of algae makes it possible to obtain synthetic gas of various compositions, which consists of hydrogen, methane, carbon monoxide, etc.

Nurcahyani et al. studied synthetic fuel production using supercritical water gasification of algae (*Chlorella vulgaris*) [19]. The biomass from algae was gasified in supercritical water at 600° and 25 MPa. The authors showed that synthesis gas after gasifier mainly consists of hydrogen, carbon monoxide with low concentrations of methane, carbon dioxide, ethylene, and ethane. The synthesis gas composition depends on operating conditions such as temperature, pressure, and residence time. For example, at a residence time of 7 s, the synthesis gas consists of hydrogen (37%), carbon monoxide (34%), methane (11%), carbon dioxide (10%), ethane (4%), ethylene (4%). The authors consider the synthesis gas as hydrogen-rich fuel.

Raheem et al. investigated the catalytic and non-catalytic steam gasification of algae (*Spirulina platensis*). CaO and Ni with Y_2O_3 support were used as the catalysts. The authors investigated the effect of temperature on synthesis gas yield. In the temperature range from 50 to 800 °C, the maximum synthesis gas yield was observed at 400–450 °C. The maximum hydrogen concentration in synthesis gas is about 43%. Steam as a gasifying agent promotes gasification reactions and water-gas shift reaction, thereby resulting in a higher hydrogen content in the product gas.

Figueira et al. presented the results of thermogravimetric analysis of the gasification of microalgae *Chlorella vulgaris* [20]. The authors showed that at 850 °C of 1 bar the algae conversion is 83.7% and synthesis gas composition mainly consists of methane (20.58%), hydrogen (16.79%), and carbon monoxide (50.14%). An increase in the gasification temperature leads to an increase in methane concentration.

Sanchez-Silva et al. studied pyrolysis, combustion and gasification characteristics of *Nannochloropsis gaditana* microalgae [21]. The maximum synthesis gas yield was observed at 400 °C. The catalytic gasification gives synthesis gas with high hydrogen concentration. For high residence time, hydrogen concentration in synthesis gas is more than 50%.

Synthetic gas as a fuel has calorimetric and emission characteristics similar to natural gas [22]. In addition, there is sufficient information in the literature that combustion of a hydrogen-rich gas leads to a reduction in NO_x emissions [23–25].

Therefore, a chain of processes: capturing CO_2 in water - processing CO_2 by photosynthesis - biomass production - gasification of biomass - use of synthesis gas as a fuel, can ve considered as the chain of hydrocarbon fuel regeneration. This chain of processes can be called photochemical hydrocarbon fuel regeneration. As was mentioned above, the gasification process is occurring at high and moderately high temperatures. In this paper, the exhaust heat after a gas turbine is suggested to use for the gasification process. The main goal of this study is to carry out a thermodynamic analysis of the process of photochemical hydrocarbon fuel regeneration, as well as to analyze the material and heat balance of the process.

Photochemical regeneration concept

The main concept of photochemical hydrocarbon fuel regeneration is based on CO_2 capture by algae, biomass gasification, and use of the gasification products as a fuel. Fig. 1 shows the schematic diagram of the photochemical hydrocarbon fuel regeneration concept. The main elements of the schematic diagram are bioreactor (CO_2 capture and algae growing) and gasifying reactor (biomass gasification).

A bioreactor is an artificial reservoir that receives moderately cooled exhaust gases. The bioreactor is surrounded by atmospheric air, the volume of water in the bioreactor is large enough. Therefore, there is no significant change in temperature in the bioreactor due to the addition of exhaust gases. Carbon dioxide and steam from combustion products dissolve in water. Algae grow in the bioreactor. Photosynthesis occurs in algae under the influence of solar radiation. Carbon dioxide and water are the main components of photosynthesis. Part of the undissolved CO_2 is released into the atmosphere, while part of the atmospheric CO_2 is dissolved in the bioreactor.

There are various types of bioreactors. For goals of photochemical regeneration, it is important to get a high residence time or contact time for CO_2 from the exhaust gases. Therefore, a vertical bioreactor with a bottom supply of exhaust gas is one of the most suitable solutions. Fig. 2 shows a vertical panel bioreactors for photochemical hydrocarbon fuel regeneration [26].

After the bioreactor, the algae are fed for gasification. Gasification is an endothermic process. Therefore, in the presented schematic diagram, it is proposed to use exhaust gases after a gas turbine. Exhaust gases after the gas turbine have a temperature from 300-400 °C to 800–900 °C and higher. In this scheme, steam gasification of biomass is proposed. Therefore, wet algae from the bioreactor can be used for further proceeding.

Wet biomass after the bioreactor is heated by exhaust gas (Exh_c) in a biomass heater. Biomass heating makes it possible to cool exhaust gas to a temperature less than 100 °C. A part of water from biomass is evaporated in the biomass heater. The mixture of hot biomass and steam is fed to a gasifying reactor. Pure steam is also fed to the gasifying reactor. This pure steam is produced from liquid water in a steam generator by use of exhaust heat (Exh_b).

The gasification process is endothermic; therefore, an external heat supply is required. Exhaust gases can be such a source of external heat. Gasification products - synthesis gas is used as a fuel, for example, together with natural gas.

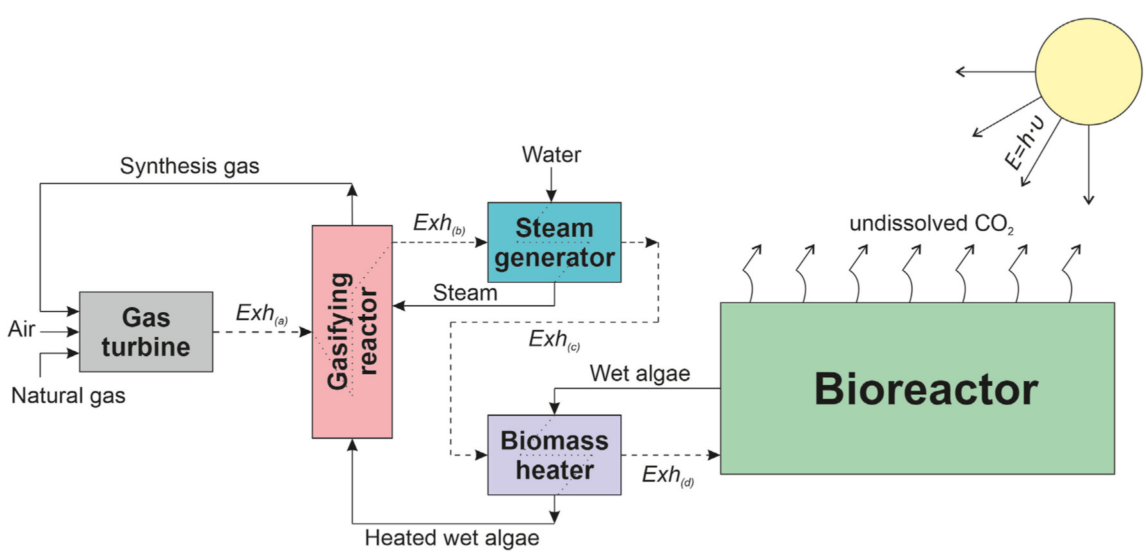

Fig. 1 – A photochemical hydrocarbon fuel regeneration concept.

Fig. 2 – A vertical panel bioreactors for photochemical hydrocarbon fuel regeneration [26].

Therefore, synthesis gas can replace the part (or all) of natural gas as a fuel.

The schematic diagram in Fig. 1 shows not only the concept of photochemical hydrocarbon fuel regeneration. The exhaust heat recuperation process is also taking a place in this scheme. Such an exhaust heat recuperation process is called thermochemical recuperation. The detailed description of the thermochemical recuperation concept is presented in the author's previous publications [27,28]. The main concept of thermochemical exhaust heat recuperation is the use of exhaust heat for the endothermic reaction of organic fuel reforming/gasification. The products of reforming/gasification have a low heating value (LHV) higher than initial organic fuel (in comparison to 1 kg of initial fuel). For example, thermochemical exhaust heat recuperation by coal steam gasification gives the synthesis gas with the heat of combustion about 1.5 times higher than the heat of combustion of 1 kg of initial coal.

Thermodynamic analysis

Gasification

The thermodynamic analysis of photochemical hydrocarbon fuel regeneration is based on the first law energy analysis and some assumptions. The thermodynamic analysis makes it possible to determine the parts of heat balance for the schematic diagram in Fig. 1. One of the main issues in the thermodynamic analysis is the determination of the synthesis gas composition after the gasifying reactor. Moreover, the thermodynamic analysis with some assumptions makes it possible to calculate mass flows in the schematic diagram in Fig. 1.

The thermodynamic equilibrium analysis was performed via Aspen HYSYS software. Aspen HYSYS is a powerful simulator for modeling steady-state processes. For the chemical process (gasification) this software is capable to model multiphase and multicomponent systems at equilibrium. To determine the equilibrium composition of a chemical process, Aspen HYSYS suggested using the Gibbs reactor. The equilibrium composition of synthesis gas is determined under the conditions of minimization of Gibbs free energy. The algorithm of thermodynamic calculation of synthesis gas composition after gasification process can be found in the author's previous work [27].

The calculation scheme for the thermodynamic analysis realized via Aspen HYSYS is presented in Fig. 3. In this calculation scheme, the exhaust gases after a gas turbine are used as an example. Moreover, to show the effect of replacing a part of natural gas with synthesis gas, the net power of a gas turbine was compared.

The thermodynamic analysis makes it possible the calculation of the equilibrium concentration of the gasification products at the given operating parameters such as temperature, inlet composition and pressure. The main concept of the Gibbs free energy minimization method is the fact the chemical system is thermodynamically favorable when its total Gibbs free energy is minimum and its differential is equal to zero [29,30]:

$$dG^t = 0 \tag{1}$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the gasification process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \tag{2}$$

where n_i is number of ith component moles; μ_i is chemical potential of ith component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G^0_{fi} + RT\ln\left(\frac{f_i}{f^0_i}\right) \tag{3}$$

where G^0_{fi} is standard Gibbs free energy of ith component formation; R is molar gas constant; T is system temperature; f_i

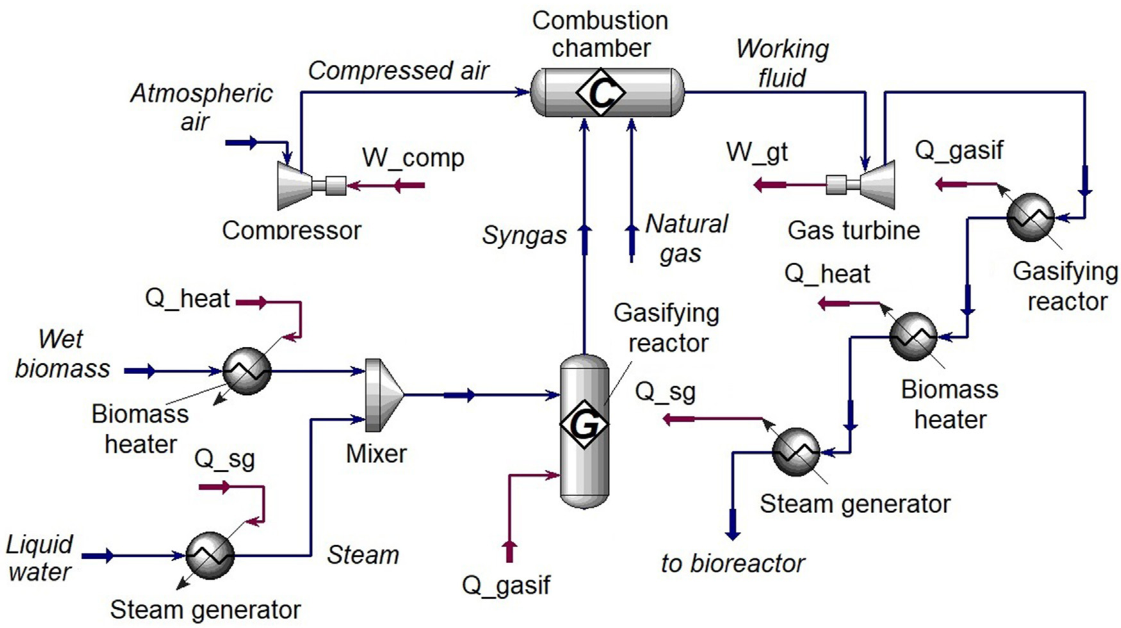

Fig. 3 – The calculation scheme for the thermodynamic analysis realized via Aspen HYSYS.

is fugacity of a pure ith component; f_i^0 is fugacity of a pure ith component at the standard state.

If the equations presented above will be combined, the equation for total Gibbs free energy of the chemical system can be written as the following:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \quad (4)$$

Assumptions

The dissolution of CO_2 in water is a well-studied process [31,32]. The solubility of CO_2 in water depends on technological and design parameters. The technological parameters include temperature, pressure, water composition, residence time, etc. The design parameters are determined by the design of the dissolution reservoir, artificial turbulators, etc. There is sufficient information in the literature that under the right cultivation conditions, the degree of carbon dioxide capture by algae can reach 99%, and the slow supply of exhaust gases causes an increase in the growth rate of algae [17].

In the present study, the determination of the solubility of CO_2 was carried out based on literature data. According to work [18], 10–30% of CO_2 from flue gases is dissolved in water. Undissolved 70–90% injects back to the atmosphere. However, in addition to the forced dissolution of CO_2 from flue gases in water, there is a natural dissolution of CO_2 from the atmospheric air. Bhola et al. showed that 4000 m^3 of water can absorb about 2200 tons of CO_2 from atmospheric air per year under normal conditions [15]. Alami et al. reported that the growth rate of biomass (algae *Spirulina*) corresponds to the level of carbon dioxide absorption of about 99%. The biomass gain mainly depends on the size of the bioreactor reservoir. Therefore, the study of CO_2 capture by water is an urgent task that can be solved experimentally under given technological and design parameters.

In this paper, it is assumed that the biomass growth corresponds up to 30% of the total capture of CO_2 from the flue gases. In other words, it is assumed that 10, 20, 30% of CO_2 from flue gases is recycled by algae. The remaining CO_2 from the flue gases is released into the atmosphere. This condition makes it possible to count only dissolved CO_2 from flue gases and ignore the background solubility of atmospheric CO_2.

The growth rate of algae also depends on technological and design factors. In addition, the type of algae determines biomass growth. In this work, *Chlorella vulgaris* is selected as the source of biomass. The growth rate of biomass is 1.88 kg of Chlorella per 1 kg of CO_2.

In addition, the following assumption are taken:

- 1 kg of dry biomass is gasified with 1.2 kg of steam;
- 1 kg of wet biomass is containing 50% of dry biomass and 50% of water;
- 0.2 kg of steam is fed to a gasifying reactor;
- steam temperature is 250 °C at pressure of 10 bar.

The exhaust gas temperature is set in the temperature range of 400 … 800 °C. The pressure of the gasification process is 1; 5; 10 bar. The mass flow rate of natural gas is 10 kg/h.

The composition of dry biomass is presented in Table 1.

Fuel and heat regeneration

According to the diagram in Fig. 3, synthesis gas after the gasification reactor partially replaces natural gas. The mass flow rate of initial methane is 10 kg/h. The lower heating value of methane is 50 MJ/kg. The thermodynamic calculations

Table 1 – The composition of dry biomass [33].

Dry biomass composition (wt%)	%
C	51.9
H	7.1
N	9.6
S	0.9
O	30.5

make it possible to calculate the degree of fuel regeneration. The ratio between the heat of combustion of produced synthesis gas and the heat of combustion of initial fuel is the fuel regeneration degree (FRD).

$$R_f = \frac{HC_{syngas}}{HC_{original}} \quad (5)$$

where HC_{syngas} is the total heat of combustion of synthesis gas that produced from captured CO_2; $HC_{original}$ is the total heat of combustion of original fuel.

To evaluate the effectiveness of heat regeneration in the photochemical fuel regeneration system, the heat regeneration degree (HRD) was determined. The heat regeneration degree shows the ratio between regenerated heat and exhaust heat. The schematic diagrams in Fig. 1 shows that heat regeneration is occurred in tree elements – a gasifying reactor, a biomass heater, a steam generator. Therefore, HRD can be determined as follows:

$$R_h = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{gas} + H_{rec}^{st.gen}}{H_{exh}}, \quad (6)$$

where H_{rec} – a regenerated heat; $H_{rec}^{gas}, H_{rec}^{st.gen}$ – a regenerated heat in the gasifying reactor, the biomass heater, the steam generator, respectively; H_{exh} – an exhaust heat, kJ.

The regenerated heat in the gasifying reactor is the enthalpy of the gasification process. Therefore, the recuperated heat in the gasifier can be determined as follows:

$$H_{rec}^{gas} = \Delta H_{298} + \Delta H_{sg}, \quad (7)$$

where H_{reac} – enthalpy of the algae gasification; ΔH_{sg} – increasing enthalpy of synthesis gas in a gasifier due to heating, kJ.

The regenerated heat in the biomass heater is determined by change in enthalpy of steam-algae mixture:

$$H_{rec}^{heater} = \Delta H_{mix}, \quad (8)$$

where ΔH_{mix} – increasing enthalpy of the steam-algae mixture.

The regenerated heat in the steam generator($H_{rec}^{st.gen}$) can be calculated as follows:

$$H_{rec}^{st.gen} = \Delta Q_{st.gen} = (h_{steam} - c_p \cdot t_w^{in}) G_{steam}^{in}, \quad (9)$$

where h_{steam} is the enthalpy of steam before the gasifying reactor; G_{steam}^{in} is the mass of water for steam generation; t_w^{in} is feed water temperature; c_p is the heat capacity of water.

To calculate HRD by eq. (6), syngas composition after the gasifier as well as the enthalpy of the gasification process should be determined.

Results and discussion

Synthesis gas composition

The major species of synthesis gas of the biomass gasification process are H2, CO, CO_2, H2O, CH4 based on experimental results. There are small amounts of other gases such as ethane, ethylene, etc, but these components are mainly produced due to the high-pressure gasification process. In this paper, the gasification pressure is in the range of up to 10 bar. Various chemical reactions occur in the gasifying reactor. The list of major reactions which have a noticeable contribution to the material and heat balance is presented in Table 2. Other chemical reactions may occur in the reformer, which is a combination of the reactions shown in Table 2.

It is clear that among several chemical reactions from Table 2, the most favorable reactions are the reactions with the minimum value in the Gibbs free energy. In this paper, the FactSage online database is utilized to calculate the change in the Gibbs free energy [34]. The change in the Gibbs free energy as a function of temperature for the chemical reactions from Table 2 is presented in Fig. 4. The reactions in Fig. 4 are numbered in the same way as in Table 2.

Fig. 4 shows that carbon gasification reactions (6 and 7 from Table 2) are most favorable in the temperature range up to 600 °C. In the temperature range up to 600 °C the forward water-gas shift reaction (2, see Table 2) is also favorable. The behaviors of the reactions (2, 6, 7) can explain the production

Table 2 – Possible chemical reactions in the gasifying reactor.

No.	Reaction	ΔH_{298}, (kJ/mol)
1	$CH_4 + H_2O \Leftrightarrow CO + 3H_2$	206.4
2	$CO + H_2O \Leftrightarrow CO_2 + H_2$	−41.1
3	$CH_4 + CO_2 \Leftrightarrow 2H_2 + 2CO$	247.3
4	$C + 2H_2 \Leftrightarrow CH_4$	- 74.8
5	$C + CO_2 \Leftrightarrow 2CO$	−173.3
6	$C + H_2O \Leftrightarrow CO + H_2$	90.13
7	$C + 2H_2O \Leftrightarrow CO_2 + 2H_2$	131.3

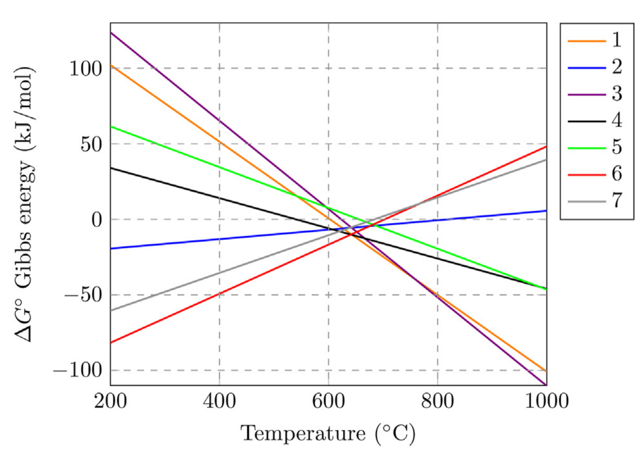

Fig. 4 – The Gibbs free energy as a function of temperature for reactions in Table 2.

of synthesis gas enriched with hydrogen. However, Fig. 4 also shows that in the temperature range up to 600 °C, hydrogen and carbon monoxide can react to produce methane. The backward reactions of methane reforming (1 and 3, see Table 2) are occurring in the temperature range up to 600 °C.

The operating parameters (temperature and pressure) have a significant effect on the composition of the synthesis gas after the gasification of algae. The composition of synthesis gas for various operating parameters (temperature and pressure) is shown in Fig. 5. Fig. 5 shows that the gasification products contain combustible components (H_2, CO, CH_4) and non-combustible components (H_2O, CO_2, N2). Methane is produced mainly by the backward reactions of methane reforming. Hydrogen and carbon monoxide is the main products of the gasification reactions. Steam in the synthesis gas is unreacted water because for 1 kg of algae 1.2 kg is water is used for the gasification process. Adding more than a stoichiometric amount of water to the reaction mixture makes it possible to minimize the formation of free carbon in the synthesis gas. Therefore, it is advisable from a practical point of view to add an excess amount of water.

In addition, Fig. 5 shows that with increasing temperature, the mole fraction of methane decreases. This fact is in good agreement with Fig. 4. With an increase in temperature, the equilibrium of the reactions of methane reforming with steam and carbon dioxide shifts in the direction of methane consumption. Fig. 5 also shows that an increase in temperature leads to a decrease in the concentration of carbon dioxide. This is due to an increase in temperature, the equilibrium of the water-gas shift reaction is moved to the left, and the Boudouard reaction (5, see Table 2) is shifted towards the formation of carbon monoxide.

An increase in temperature led to an increase in the mole fraction of hydrogen and carbon monoxide. However, an increase in pressure leads to a decrease in the mole fraction of hydrogen and carbon monoxide. Le-Chatelier's principle provides an explanation for this fact. The gasification products have a higher number of moles of gaseous substances than the initial components. Therefore, increasing pressure shifts the equilibrium to the left. On the other hand, an increase in pressure leads to an increase in the mole fraction of methane. This is due to the fact that the cracking reaction of 1 mol of methane gives 2 mol of hydrogen. Therefore, with increasing pressure, the equilibrium of the reaction shifts towards the formation of methane.

Heat balance

The operating parameters have a significant impact on the synthesis gas composition after algae gasification. Therefore, the operating parameters have a significant impact on the enthalpy of the gasification process. The main gasification reactions from Table 2 are endothermic. Therefore, external heat is required for algae gasification. Fig. 6a shows the enthalpy of the gasification process as a function of temperature for various pressures.

Fig. 6a shows that with an increase in temperature, the enthalpy of the process increases and reaches a value close to the maximum (about 9 MJ/kg_{algae}) at temperatures above 900 °C. On the other hand, the pressure has no significant effect on the enthalpy of gasification. An increase in pressure leads to a slight increase in the enthalpy of the process. The gasification enthalpy increases significantly in the temperature range from 400 to 700 °C. In the temperature range above 700 °C, there is a slight increase in the enthalpy of gasification with increasing temperature.

The enthalpy of the process determines the amount of external heat that must be supplied to the gasification reactor (according to the diagram in Fig. 1). In addition, the enthalpy of the process shows how much the heat of combustion of synthesis gas increases in comparison with the heat of combustion of algae according to the first law energy conservation equation.

$$LHV_{synfuel} = LHV_{fuel} + \Delta H_{298}, \tag{10}$$

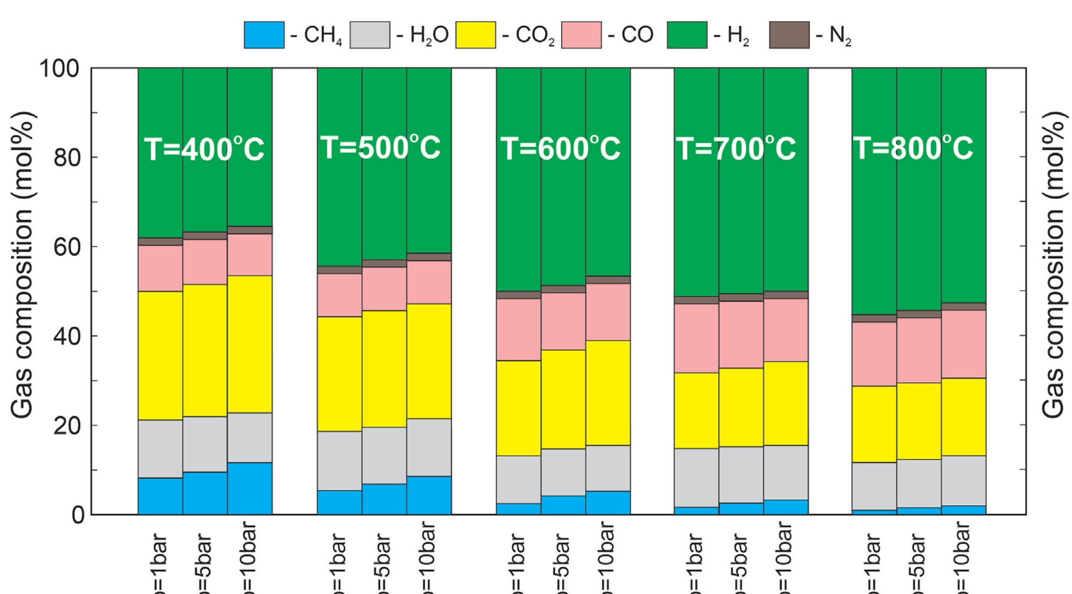

Fig. 5 – Synthesis gas composition for various operating parameters such as temperature and pressure.

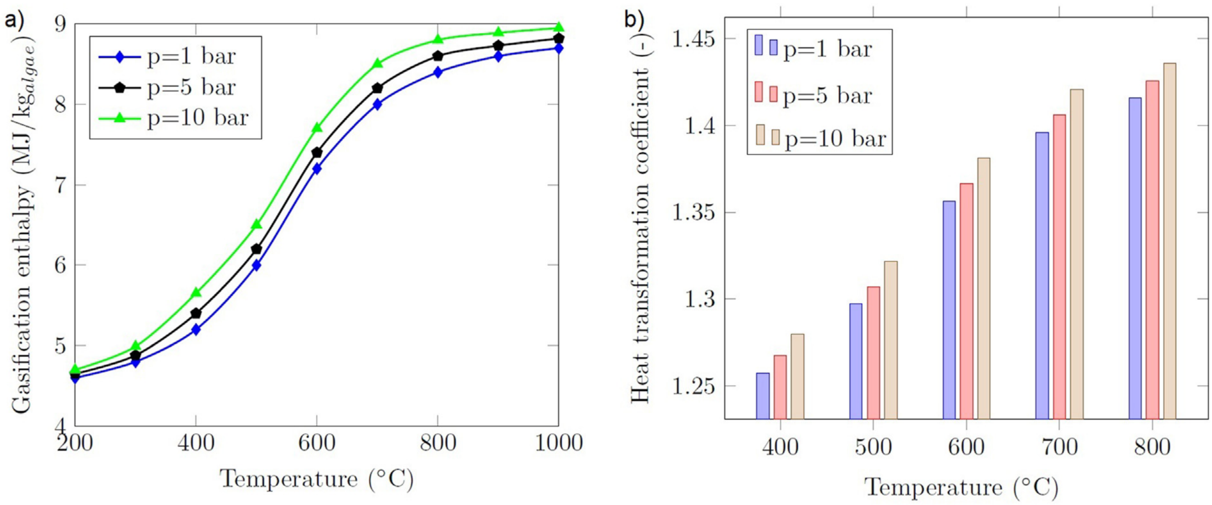

Fig. 6 – Enthalpy of algae gasification (a) and heat transformation coefficient for various operating conditions.

where *LHVsynfuel*, LHV_{algae} – low heating value of synthetic fuel and initial fuel, respectively; ΔH_{298} – enthalpy of the gasification process.

One of the important criteria for the efficiency of the photochemical regeneration system is the heat transformation coefficient in the gasification reactor. The ratio between the lower heating value (LHV) of synthetic fuel to the lower heating value of initial algae biomass is named the heat transformation coefficient (HTC). HTC can be calculated as follows:

$$n_q = LHV_{synfuel}/LHV_{algae}. \quad (11)$$

Fig. 6b shows the heat transformation coefficient for various operating parameters. Fig. 6b makes it clear that the gasification of the algae provides an increase in the heat of combustion up to 43%. At the same time, it should be noted that waste heat is used for gasification (according to the diagram in Fig. 1). Therefore, no additional heat input is required.

In addition, it should be noted that 1.2 kg of water is used to gasify 1 kg of algae. Therefore, the combustion products of synthesis gas will contain an additional 1.2 kg of water. The condensation of this water will provide an additional 2709.6 kJ of heat. Therefore, HTC based on higher heating value (HHV) will be about 12% higher than HTC based on LHV.

According to the schematic diagram in Fig. 3, synthesis gas after the gasification reactor partially replaces natural gas. The mass flow rate of initial methane is 10 kg/h. The lower heating value of methane is 50 MJ/kg. The thermodynamic calculations make it possible to calculate the degree of fuel regeneration. The total heat of combustion for the scheme in Fig. 3 of original fuel is 500 MJ/h (10 kg/h of methane with 50 MJ/kg). Table 3 present the results of heat and mass balance compilation. Table 3 shows that the degree of fuel regeneration at 30% of captured CO_2 from flue gases is 0.8 … 0.9. An increase in captured CO_2 in water leads to an increase in the degree of fuel regeneration. If 36% of CO_2 is captured from flue gases, then the degree of fuel regeneration is about unity (at a temperature of 600 °C). Therefore, to improve the efficiency of photochemical fuel regeneration systems, CO_2 capture and the rate of algae growth are important factors.

In the schematic diagram from Fig. 1, exhaust heat is used for an algae gasification process. Therefore, a part of exhaust heat is regenerated to the fuel cycle of a gas turbine. The heat regeneration degree (HRD) is one of the main criteria for the efficiency of the heat regeneration systems. Fig. 7 presents HRD as a functions of temperature and pressure.

Fig. 7 shows that the heat regeneration degree strongly depends on temperature and slightly depends on pressure. In the temperature range up to 600 °C, HRD increases because the enthalpy of the gasification process is increasing. The maximum value of HRD is achieved at 600 °C. In the temperature range above 600 °C, HRD is slightly decreasing. This is

Table 3 – Heat and mass balance for the hydrocarbon fuel regeneration system.

Temperature, °C	Share of captured CO_2	Mass of captured CO_2, kg/h	Syngas heat of combustion, MJ/h	Fuel regene-ration degree, [–]
400	10%	2.7225	136.5051	0.27301
	20%	5.4450	273.0101	0.54602
	30%	8.1675	409.5152	0.81903
600	10%	2.7225	147.2535	0.29450
	20%	5.4450	294.5070	0.58901
	30%	8.1675	441.7605	0.88352
800	10%	2.7225	152.6277	0.30525
	20%	5.4450	305.2554	0.61051
	30%	8.1675	457.8831	0.91576

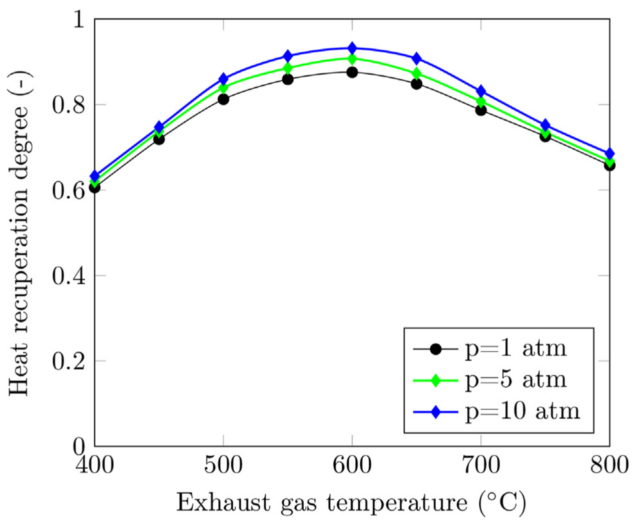

Fig. 7 – Heat regeneration degree in the photochemical fuel regeneration system.

due to fact that the enthalpy of the gasification process is reached almost its maximum value but the denominator in eq. (6) is increasing with an increase in temperature. Fig. 7 depicts that the photochemical fuel regeneration system from Fig. 1 is recovering about 90% of exhaust heat.

After analyzing the schematic diagram it can be concluded that the photochemical fuel regeneration system can regenerate up to 90% of hydrocarbon fuel (methane) when 30% of CO_2 from the flue gases is captured in water and proceeded by algae. Moreover, up to 90% of exhaust heat can be recovered to the fuel cycle in the photochemical fuel regeneration system. Therefore, the photochemical fuel regeneration system can be considered as an environmentally friendly energy-saving technology.

Conclusion

TThis paper study the photochemical hydrocarbon fuel regeneration system that converted carbon dioxide to hydrogen-rich fuel. CO_2 from the flue gases is captured by water where algae are using CO_2 for photosynthesis reaction. Algae biomass is gasified with steam to produce hydrogen-rich fuel. To compile the heat balance, the thermodynamic analysis was performed via Aspen HYSYS. Based on the thermodynamic analysis, the enthalpy of gasification and the composition of gasification products was determined.

With an increase in temperature, the mole fraction of hydrogen and carbon monoxide is increasing while the mole fraction of methane is decreasing. The enthalpy of gasification is increasing with an increase in temperature. The enthalpy of gasification is about 5 MJ/kg_{algae} at 400 °C and 8.7 MJ/kg at 800 °C. The pressure has no significant impact on the composition of gasification products and the enthalpy of gasification. To understand the calorimetric performance of the gasification products, the heat transformation coefficient (HTC) is determined. HTC is increasing with an increase in temperature. The lower heating value of the gasification products is 1.43 times higher (at 800 °C) than the lower heating value of original biomass. Based on the initial parameters, the fuel regeneration degree was calculated. The degree of fuel regeneration at 30% of captured CO_2 from flue gases is 0.8 … 0.9. An increase in captured CO_2 in water leads to an increase in the degree of fuel regeneration. If 36% of CO_2 is captured from flue gases, then the degree of fuel regeneration is about unity (at a temperature of 600 °C).

In the presented photochemical hydrocarbon fuel regeneration concept, external heat for algae biomass gasification is used from an exhaust gas after fuel-consuming equipment. Therefore, the gasification part of the photochemical fuel regeneration system provides exhaust heat regeneration. The heat regeneration degree was calculated. The maximum heat regeneration degree is about 0.9 at the temperature of 600 °C. Therefore, the photochemical fuel regeneration system is recovered about 90% of exhaust heat. The photochemical fuel regeneration system can be considered as an environmentally friendly energy-saving technology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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REFERENCES

[1] International Energy Agency. World energy balances: overview. International Energy Agency Paris; 2020.
[2] Pietzcker Robert C, Osorio Sebastian, Rodrigues Renato. Tightening eu ets targets in line with the European green deal: impacts on the decarbonization of the eu power sector. Appl Energy 2021;293:116914.
[3] Luo Shihua, Hu Weihao, Wen Liu, Xu Xiao, Huang Qi, Chen Zhe, Lund Henrik. Transition pathways towards a deep decarbonization energy system—a case study in sichuan, China. Appl Energy 2021;302:117507.
[4] Chen Xu, Li Zhongshu, Gallagher Kevin P, Mauzerall Denise L. Financing carbon lock-in in developing countries: bilateral financing for power generation technologies from China, Japan, and the United States. Appl Energy 2021;300:117318.
[5] Danaci David, Bui Mai, Petit Camille, Niall Mac Dowell. En route to zero emissions for power and industry with amine-based post-combustion capture. Environ Sci Technol 2021;55(15):10619–32.
[6] Romeo Luis M, Minguell Diego, Shirmohammadi Reza, Andres Josee M. Comparative analysis of the efficiency penalty in power plants of different amine-based solvents for co2 capture. Ind Eng Chem Res 2020;59(21):10082–92.
[7] Hernández Elisa, Santiago Rubén, Moya Cristian, Vela Sonia, Navarro Pablo, Palomar José. Close-cycle process to produce co2-derived propylene carbonate based on amino acid catalyst and water. J CO2 Util 2021;52:101656.
[8] Kofi Ayittey Foster, Ann Obek Christine, Saptoro Agus, Kumar Perumal, Wong Mee Kee. Process modifications for a

hot potassium carbonate-based co2 capture system: a comparative study. Greenh GasesSci Technol 2020;10(1):130–46.
[9] Siddiqui Osamah, Ishaq Haris, Chehade Ghassan, Dincer Ibrahim. Performance investigation of a new renewable energy-based carbon dioxide capturing system with aqueous ammonia. Int J Energy Res 2020;44(3):2252–63.
[10] Nguyen Hoan Le Quoc, Wong David Shan-Hill. Integration of rich and lean vapor recompression configurations for aqueous ammonia-based co2 capture process. Chem Eng Res Des 2021;169:86–96.
[11] Zhu Bin, Jiang Xu, He Shanshan, Yang Xiaobin, Long Jun, Zhang Yanqiu, Lu Shao. Rational design of poly (ethylene oxide) based membranes for sustainable co 2 capture. J Mater Chem 2020;8(46):24233–52.
[12] Chernikova Valeriya, Shekhah Osama, Belmabkhout Youssef, Mohamed Eddaoudi. Nanoporous fluorinated metal–organic framework-based membranes for co2 capture. ACS Appl Nano Mater 2020;3(7):6432–9.
[13] Alalwan Hayder A, Alminshid Alaa H. Co2 capturing methods: chemical looping combustion (clc) as a promising technique. Sci Total Environ 2021:147850.
[14] Kuo Po-Chih, Illathukandy Biju, Wu Wei, Chang Jo-Shu. Energy, exergy, and environmental analyses of renewable hydrogen production through plasma gasification of microalgal biomass. Energy 2021;223:120025.
[15] Bhola V, Swalaha F, Ranjith Kumar R, Singh M, Bux F. Overview of the potential of microalgae for co 2 sequestration. Int J Environ Sci Technol 2014;11(7):2103–18.
[16] Iglina Tatyana, Iglin Pavel, Pashchenko Dmitry. Industrial co2 capture by algae: a review and recent advances. Sustainability 2022;14(7):3801.
[17] Alami Abdul Hai, Alasad Shamma, Ali Mennatalah, Alshamsi Maitha. Investigating algae for co2 capture and accumulation and simultaneous production of biomass for biodiesel production. Sci Total Environ 2021;759:143529.
[18] Deprá Mariany C, Dias Rosangela R, Severo Ihana A, de Menezes Cristiano R, Zepka Leila Q, Jacob-Lopes Eduardo. Carbon dioxide capture and use in photobioreactors: the role of the carbon dioxide loads in the carbon footprint. Bioresour Technol 2020;314:123745.
[19] Nurcahyani Puji Rahmawati, Hashimoto Soshi, Matsumura Yukihiko. Supercritical water gasification of microalgae with and without oil extraction. J Supercrit Fluids 2020;165:104936.
[20] Figueira Camila Emilia, Moreira Jr Paulo Firmino, Giudici Reinaldo. Thermogravimetric analysis of the gasification of microalgae chlorella vulgaris. Bioresour Technol 2015;198:717–24.
[21] Sanchez-Silva L, López-González D, Garcia-Minguillan AM, Valverde JL. Pyrolysis, combustion and gasification characteristics of nannochloropsis gaditana microalgae. Bioresour Technol 2013;130:321–31.
[22] Karpilov Igor, Pashchenko Dmitry. Effect of methane reforming before combustion on emission and calorimetric characteristics of its combustion process. Int J Hydrogen Energy 2021;46(76):38106–18.
[23] Lopez-Ruiz G, Fernandez-Akarregi AR, Diaz L, Urresti I, Alava I, Blanco JM. Numerical study of a laminar hydrogen diffusion flame based on the non-premixed finite-rate chemistry model; thermal nox assessment. Int J Hydrogen Energy 2019;44(36):20426–39.
[24] Habib Mohamed A, Mokheimer Esmail MA, Sanusi Sofihullahi Y, Nemitallah Medhat A. Numerical investigations of combustion and emissions of syngas as compared to methane in a 200 mw package boiler. Energy Convers Manag 2014;83:296–305.
[25] Scribano Gianfranco, Cheng Xinwei, Tran Manh-Vu. Numerical simulation of the effects of hydrogen and carbon monoxide ratios on the combustion and emissions for syngas fuels in a radiant burner. Energy 2021;214:118910.
[26] Lehr Florian, Posten Clemens. Closed photo-bioreactors as tools for biofuel production. Curr Opin Biotechnol 2009;20(3):280–5.
[27] Pashchenko Dmitry. Industrial furnaces with thermochemical waste-heat recuperation by coal gasification. Energy 2021;221:119864.
[28] Pashchenko Dmitry. Thermochemical waste-heat recuperation as on-board hydrogen production technology. Int J Hydrogen Energy 2021;46(57):28961–8.
[29] Aktaş Seda, Karakaya Mustafa, Avcı Ahmet K. Thermodynamic analysis of steam assisted conversions of bio-oil components to synthesis gas. Int J Hydrogen Energy 2009;34(4):1752–9.
[30] Özkara-Aydınoğlu Şeyma. Thermodynamic equilibrium analysis of combined carbon dioxide reforming with steam reforming of methane to synthesis gas. Int J Hydrogen Energy 2010;35(23):12821–8.
[31] Michaelides Efstathios E. Thermodynamic analysis and power requirements of co2 capture, transportation, and storage in the ocean. Energy 2021;230:120804.
[32] Thomas Carelle, Dehaeck Sam, De Wit Anne. Convective dissolution of co2 in water and salt solutions. Int J Greenh Gas Control 2018;72:105–16.
[33] Yu Hong, Chen Wanru, Luo Xiang, Pang Chengheng, Lester Edward, Wu Tao. Microwave-enhanced pyrolysis of macroalgae and microalgae for syngas production. Bioresour Technol 2017;237:47–56.
[34] Bale Ch W, Chartrand P, Degterov SA, Eriksson G, Hack K, Mahfoud R Ben, Melançon J, Pelton AD, Petersen S. Factsage thermochemical software and databases. Calphad 2002;26(2):189–228.